Problem 3.3.3. The goal of this problem is to give an alternative proof of Theorem 3.3.1, not using any of the previous results of Chapter 3.

Let A_1, A_2, ..., A_n be n algebras with units 1_1, 1_2, ..., 1_n, respectively. Let $A = A_1 \oplus A_2 \oplus \cdots \oplus A_n$. Clearly, $1_i 1_j = \delta_{ij} 1_i$, and the unit of A is $1 = 1_1 + 1_2 + \cdots + 1_n$.

For every representation V of A, it is easy to see that $1_i V$ is a representation of A_i for every $i \in \{1, 2, \ldots, n\}$. Conversely, if V_1, V_2, ..., V_n are representations of A_1, A_2, ..., A_n, respectively, then $V_1 \oplus V_2 \oplus \cdots \oplus V_n$ canonically becomes a representation of A (with $(a_1, a_2, \ldots, a_n) \in A$ acting on $V_1 \oplus V_2 \oplus \cdots \oplus V_n$ as $(v_1, v_2, \ldots, v_n) \mapsto (a_1 v_1, a_2 v_2, \ldots, a_n v_n)$).

(a) Show that a representation V of A is irreducible if and only if $1_i V$ is an irreducible representation of A_i for exactly one $i \in \{1, 2, \ldots, n\}$, while $1_i V = 0$ for all the other i. Thus, classify the irreducible representations of A in terms of those of A_1, A_2, ..., A_n.

(b) Let $d \in \mathbb{N}$. Show that the only irreducible representation of $\operatorname{Mat}_d(k)$ is k^d, and every finite dimensional representation of $\operatorname{Mat}_d(k)$ is a direct sum of copies of k^d.

Hint: For every $(i, j) \in \{1, 2, \ldots, d\}^2$, let $E_{ij} \in \operatorname{Mat}_d(k)$ be the matrix with 1 in the ith row of the jth column and 0's everywhere else. Let V be a finite dimensional representation of $\operatorname{Mat}_d(k)$. Show that $V = E_{11} V \oplus E_{22} V \oplus \cdots \oplus E_{dd} V$, and that $\Phi_i : E_{11} V \to E_{ii} V$, $v \mapsto E_{i1} v$ is an isomorphism for every $i \in \{1, 2, \ldots, d\}$. For every $v \in E_{11} V$, denote $S(v) = \langle E_{11} v, E_{21} v, \ldots, E_{d1} v \rangle$. Prove that $S(v)$ is a subrepresentation of V isomorphic to k^d (as a representation of $\operatorname{Mat}_d(k)$), and that $v \in S(v)$. Conclude that $V = S(v_1) \oplus S(v_2) \oplus \cdots \oplus S(v_k)$, where $\{v_1, v_2, \ldots, v_k\}$ is a basis of $E_{11} V$.

(c) Deduce Theorem 3.3.1.

Remark 3.3.4. Here is yet another proof of Theorem 3.3.1, using Lemma 3.1.6. Let X be an n-dimensional representation of A. Let $\{x_1, ..., x_n\}$ be a basis of X. Then there is a unique homomorphism $\psi : A^n \to X$ such that $\psi(a_1, ..., a_n) = \sum_i a_i x_i$, and it is surjective. Hence X is a quotient of A^n. But we have seen that $A = \bigoplus_{i=1}^r d_i V_i$,